

the effectiveness of self-critique on computer tasks by proposing Recursively Criticizes and Improves (RCI). This method utilizes a pre-trained LLM agent to execute computer tasks, guiding it using a simple prompting scheme where the agent recursively criticizes and improves its output.

Although Reflexion [69] and RCI [36] do not explicitly utilize human feedback, Huang et al. [29] points out that these approaches employ the labels of test questions to determine when to halt the self-correction loop. Consequently, they are implicitly guided by human feedback. Huang et al. [29] categorizes these methods as *self-correction with oracle feedback* (SCO).

Due to the difficulty of LLMs relying solely on their inherent capabilities to correct responses for reasoning tasks [29], Tyen et al. [81] decomposes the self-correction process and observes challenges in accurately identifying error locations for LLMs. Consequently, they propose Backtracking (BT), which guides the correction process of models by providing the initial error location in the problem.

Furthermore, the incorporation of external tools can enhance the model’s self-correction performance. For instance, to improve the code generation capabilities of the model, Jiang et al. [34] utilizes error messages generated by an external interpreter as feedback, requiring LLM to debug the initially generated code acting as a professional programmer. Introducing external sources of knowledge, such as Wikipedia, can also significantly enhance the accuracy of the model in reasoning tasks [96, 43].

In addition to leveraging external tools, certain efforts revolve around the utilization of external models to generate feedback. Paul et al. [59] introduces a critic model that generates structured feedback for the responses produced by the LLM during the self-correction process. The feedback is then incorporated into the input, allowing the LLM to refine its own output iteratively. Likewise, Akyürek et al. [3] presents a collaborative framework involving multiple agents, where the critique generator is trained with the aim of maximizing the performance of the reasoning LLM.

B.3 Self-consistency

In our study, self-consistency represents a category of robustness methods, typically involving the generation of multiple diverse answers to a given question and subsequently converging them. This approach serves to mitigate the impact of noise inherent in the question. Among existing methods, two primary approaches are employed to generate distinct answers: (1) generating multiple samples via randomized smoothing on input, followed by separate and independent reasoning processes [62, 102, 98], and (2) leveraging the inherent randomness in the decoding process of the LLM to sample multiple distinct reasoning paths for the same input [83].

Sampling diverse outputs. Relying on the idea that an intricate reasoning problem usually allows for various distinct thinking ways, each leading to the unique correct solution, Wang et al. [83] first explores a varied array of reasoning paths and subsequently identifies the most consistent answer by majority vote. This approach boosts the model’s performance across various reasoning tasks.

Randomized smoothing over inputs. Randomized smoothing serves as a straightforward yet effective mechanism commonly employed to defend against adversarial attacks by introducing multiple perturbations to the input to smooth potential noise [13, 98, 62, 102].

To address the vulnerability of LLMs to jailbreaking attacks, Robey et al. [62] proposes Smooth LLM (SM), which is based on randomized smoothing. Utilizing the discovery that prompts generated adversarially exhibit fragility to character-level modifications, this method initiates by randomly introducing perturbations to multiple copies of a given input prompt. Subsequently, it consolidates the associated predictions to identify adversarial inputs.

However, as pointed out in [102], randomized smoothing relies on introducing additional noise to input, and its certification success hinges on the model’s performance with corrupted data. To address the issue, Zhang et al. [102] proposes Self-denoising (SD), which denoises the corrupted inputs with LLMs using the multitasking nature of LLMs.

B.4 External Supervision

In addition to self-correction and self-consistency methods, there is a line of methods that require external supervision or information. In our study, we compare diverse denoising methods that require varying degrees of external supervision, including (1) ground truth for test questions, (2) noise positions, and (3) additional clean demonstrations.

To provide a thorough comparison, We adopt Self-correction with Oracle feedback (SCO) [29] as a strong baseline, which uses the labels of test questions to determine when to terminate the

self-correction loop, as discussed in B.2. We also employ Backtracking (BT) [81], which necessitates the location of the initial noise in the prompt, to demonstrate the effectiveness of our method. In addition, we utilize Contrastive Chain-of-Thought (CC) [9] in experiments which needs similar external supervision as our proposed method: a clean demonstration. This approach enhances reasoning by presenting the LLM with all the clean and noisy examples concurrently.

B.5 Relation with Literature

There are some implicit connections with existing papers. Here, we would like to further clarify our work’s major differences and unique contributions, compared with [92], [54] and [9].

As discussed in B.1, Yao et al. [92] proposes Tree-of-Thought (ToT) for deliberate reasoning tasks. ToT decomposes a given problem into several reasoning steps and generates multiple answers for each step, ultimately constructing a tree structure. Subsequently, ToT employs BFS or DFS to traverse the tree, yielding the final rationale and answer. Namely, ToT focuses on the rationale (thoughts) generation in the output space of an LLM, wherein the generated thoughts can be irrelevant or inaccurate. By contrast, Our work focuses on the input space with noisy prompting examples. From this view of input space and output space, ToT is orthogonal to our work.

Moreover, ToT focuses on problem-solving by reasoning, *e.g.*, Game of 24 and Mini Crosswords, while our work investigates the robustness problem of noisy rationales. ToT does not consider imperfect scenarios with noisy inputs. This means that when ToT is prompted by examples with noisy rationales, the reasoning might also be degenerated. Hence, a further investigation on reasoning robustness with these up-to-date prompting techniques, possibly with a focus on the output space, can be conducted based on our work.

As for [54], this work explores the question-answer mapping of prompting examples. They find that breaking the mapping has nearly no influence on LLM’s in-context learning.

However, it’s important to note that [54] primarily focuses on the relationship between input and label, investigating the question-answer mapping of prompting examples. Their work does not delve into the role of rationales in the examples, which is the central focus of our research. In contrast, our experiments keep the input-label mapping intact and instead explore the impact of introducing noise into the rationales. This allows us to isolate the effect of rationale quality on the model’s performance without altering the fundamental relationship between questions and answers.

Hence, when the form of prompting example shifts from (question, answer) to (question, rationale, answer), more aspects can be investigated. We conduct additional experiments to further explore the mapping among question, rationale, and answer in Sec. 4.

Chia et al. [9] proposes Contrastive Chain-of-Thought (CC) [9] to enhance LLM reasoning. In our work, we focus on improving reasoning in noisy settings and define the noisy rationale problem as “factually inaccurate or irrelevant reasoning steps paired with valid question-answer prompts.” Fig. 1 shows an example. Here, only one rationale is given in each demonstration, which can be potentially noisy but unknown to the model. However, in CC’s setting, each ICL demonstration explicitly includes a clean rationale and a wrong rationale. Notably, this rationale is wrong instead of noisy as it induces the wrong answer. Therefore, CC’s setting and ours are totally different. In addition, empirical results in Tab. 8 show that our method CD-CoT significantly outperforms CC when given the same information.

The problem of noisy rationale is challenging and non-trivial. We believe that the investigation of noisy rationale is still in the budding stage. Several potential solutions can be progressively devised to address this problem. For example, accessing external databases is a promising solution. However, this retrieve augmented generation (RAG) is usually used to address the lack of knowledge of LLMs but not for data denoising. Therein, verifying the factual relevance and accuracy of a rationale is quite challenging: (1) there is a semantic gap between a rationale and the corresponding query instructions of a database; (2) how can the models realize the need for data verification and then find a suitable database for that are also cutting-edge research directions?

Hence, we believe that RAG for denoising is not a trivial solution. More research attention and effort should be paid to achieve this solution. In this work, we verify that an LLM can achieve good denoising results when prompted with only one clean rationale and several noisy rationales. This solution, sharing a similar spirit to RAG that requires additional information, significantly mitigates the intrinsic vulnerabilities of LLMs.

C Benchmark

In this section, we provide a comprehensive overview of the NoRa benchmark, including the motivation for constructing NoRa (Appendix C.1), a further discussion on the origins and impact of noisy rationales in CoT demonstrations (Appendix C.2), real-world examples of noisy rationales from both model and human perspectives (Appendix C.3), the detailed process of generating noise for the NoRa dataset (Appendix C.4), a comprehensive datasheet for the NoRa dataset (Appendix C.5), and full examples of the NoRa dataset across various reasoning tasks (Appendix C.6).

C.1 Motivation

The robustness of LLMs against noisy inputs deserves more attention, particularly in the context of Chain-of-Thought (CoT) reasoning. While CoT methods have shown remarkable effectiveness, they face two fundamental challenges: the susceptibility to selected demonstrations and the complexity of generating high-quality demonstrations.

Recent studies have revealed that LLMs can be easily distracted by irrelevant or adversarial context, and their reasoning process is unstable under small prompt modifications. Furthermore, misleading information can cause LLMs to agree with factual errors. These vulnerabilities become particularly concerning when we consider that demonstrations in practice are not always clean or accurate.

The problem of noisy rationales in demonstrations arises from various practical scenarios:

1. **Human-generated noise:** Just as humans make mistakes in data annotation (which has motivated extensive research in label-noise learning), they can also provide imperfect demonstrations. Different users with varying expertise and background knowledge may contribute demonstrations of varying quality.
2. **System-generated noise:** Noise can emerge from automated systems, including dialogue systems and machine-generated data, especially in crowdsourced platforms.
3. **Demonstration selection challenges:** While incorporating human supervision in dealing with noisy rationales is possible, it is often expensive and impractical at scale. The sensitivity of LLMs to in-context learning examples makes this particularly challenging.

Our empirical studies have revealed widespread vulnerability among prevailing LLMs to noisy rationales, with existing reasoning methods showing limited efficacy. Compared to clean rationales, we observe performance decreases of 0.4-19.8% with irrelevant noise and more severe degradation of 2.0-66.8% with inaccurate noise.

The NoRa benchmark addresses this crucial gap in current research. While most existing work assumes clean demonstrations for in-context learning, we argue that noisy rationales represent a practical and challenging problem that requires immediate attention. This benchmark serves two primary purposes:

1. It provides a systematic way to evaluate LLM robustness against different types of noise in rationales.
2. It helps identify the limitations of current approaches and motivates the development of more robust reasoning strategies.

We believe the NoRa dataset and our findings will contribute significantly to the development of more trustworthy foundation models capable of handling real-world, imperfect demonstrations.

C.2 A Further Discussion on Noisy Rationales in CoT Demonstrations

In this part, we further clarify the background and settings of the Noisy-R problem investigated in our work, *i.e.*, noisy rationales in CoT demonstrations. Having established the presence of irrelevant and inaccurate thoughts as primary noise types, we now turn to their origins and impact within human-model interactions.

Irrelevant and inaccurate thoughts can be generated by both the model and humans. On the one hand, irrelevant or inaccurate thoughts are likely to appear within the rationales of LLMs’ answers. These rationales, drawn from historical dialogues between users and the model, can create a noisy